Explicit irrationality bounds for odd powers of π

Nested logarithmic Hermite–Padé constructions,
prime divisors, and algebraic-approximation transfer

Expository research notes

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Abstract

We develop a nested-interval version of a classical logarithmic Hermite–Padé construction and give explicit upper bounds for the irrationality exponents of odd powers of π. The construction uses Gaussian-integer polynomials in $\log \mathrm{i} = \mathrm{i}\pi/2$, rather than a direct sequence of binary linear forms in $1, \pi^k$. We prove integrality, an exact nonvanishing determinant identity, a prime-divisor estimate described by weighted arcs on a circle, and uniform analytic bounds. A minimal-polynomial resultant argument handles every rational approximation, including reducible-binomial and rational-power cases. A twelve-layer choice gives

$$\mu(\pi^3) < 1203.973, \quad \mu(\pi^5) < 42461, \quad \mu(\pi^7) < 1272103, \quad \mu(\pi^9) < 35398937.$$

The numerical inequalities are certified by ball arithmetic. Detailed background, worked examples, the order of parameter choices, and unsuccessful finite experiments are included for further investigation.

Status and scope. This is an expanded working manuscript, not a journal submission or an independently refereed paper. It makes no priority, record, or global optimality claim. The displayed upper bounds follow from the arguments given here; finite tests and numerical certificates have more limited roles, which are stated explicitly. No upper bound below 100 for $\mu(\pi^3)$ is established.

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1 Purpose, conventions, and main conclusions

1.1 What is being bounded?

For an irrational real number ξ, its irrationality exponent is

$$\mu(\xi) = \sup\left\{u > 0 : 0 < \left|\xi - \frac{p}{q}\right| < q^{-u} \text{ for infinitely many reduced } p/q,\ q > 0\right\}.$$

Thus $\mu(\xi) \le E$ means that for every $\varepsilon > 0$ there are only finitely many reduced fractions with error below $q^{-E-\varepsilon}$. Equivalently, after changing a positive constant, one has

$$\left|\xi - \frac{p}{q}\right| \ge c_\varepsilon q^{-E-\varepsilon} \tag{1.1}$$

for all p, q with $q > 0$. This does not assert the same inequality with exponent exactly E.

Dirichlet's theorem gives $\mu(\xi) \ge 2$. Obtaining an explicit upper bound for a particular transcendental number is a separate problem. The finiteness of the exponents of fixed powers of π is already accessible through classical quantitative transcendence theory. Our objective is to spell out one concrete construction and track its constants, not to claim a first finiteness theorem.

All logarithms in estimates are natural logarithms. We use

$$\lambda = \log \mathrm{i} = \frac{\mathrm{i}\pi}{2}, \qquad \Theta = \frac{\pi}{2}.$$

The symbols $o(n)$ and $O(\log n)$ refer to $n \to \infty$ with the complete interval configuration and target k fixed. The auxiliary degree m may be large, but is fixed in this limit.

1.2 Why odd powers are natural targets

The Dirichlet beta function is

$$\beta(s) = \sum_{\ell=0}^{\infty} \frac{(-1)^\ell}{(2\ell+1)^s}.$$

For $r \ge 0$,

$$\beta(2r+1) = \frac{(-1)^r E_{2r}}{4^{r+1}(2r)!}\pi^{2r+1},$$

where $\operatorname{sech} z = \sum_{j\ge 0} E_j z^j/j!$ defines the Euler numbers. In particular,

$$\beta(1) = \frac{\pi}{4}, \qquad \beta(3) = \frac{\pi^3}{32}, \qquad \beta(5) = \frac{5\pi^5}{1536}.$$

Multiplication by a nonzero rational number preserves the irrationality exponent. For a reduced multiplier a/b, reducing $ap/(bq)$ cancels a gcd dividing $|ab|$, so its denominator remains comparable to q by constants independent of p, q.

This observation does *not* deduce an upper bound for π^3 from a rational-approximation bound for π. Taking a cube root changes the approximation field. The resultant argument below accounts for that change.

1.3 A preview of the construction

There are three components:

1. Build integer polynomials $A_{h,j,n}(x)$ such that
$$R_{h,n}(x) = \sum_{j=0}^{m} A_{h,j,n}(x)(\log x)^j$$
has a high-order zero at $x = 1$.
2. Divide every coefficient by a common integer Π_n. Its exponential rate $\gamma = \lim n^{-1} \log \Pi_n$ matters; a large constant depending only on m does not change this rate.
3. Evaluate at $x = \mathrm{i}$, compare λ with a k-th root of a rational approximation to π^k, and use a nonzero resultant.

Nested intervals change both analytic bounds and arithmetic divisors. Their effects must be assessed together.

The classical logarithmic construction appears in Mignotte [1]. Weighted logarithmic Padé systems and normality are discussed in Kawashima [2]. Here the determinant and estimates for our two-sided nested system are proved directly.

1.4 General parameter theorem

Choose integers
$$L \geq 3, \quad 0 = t_0 < t_1 < \cdots < t_{J-1} < L/2, \quad \nu_j \geq 1,$$
and put
$$M = \sum_{j=0}^{J-1} \nu_j, \qquad m = M - 1 \geq 1, \qquad c_j = 1 - \frac{2t_j}{L}.$$
An interval of relative length c_j, centred at $1/2$, is used ν_j times. Let γ be the explicit prime-divisor rate defined in Section 3. Let $Y > 0$ solve
$$2\sum_j \nu_j \arctan \frac{c_j}{2Y} = \Theta. \tag{1.2}$$
Define
$$F_c(Y) = \frac{c}{2} \log \frac{1 + (2Y/c)^2}{4} + 2Y \arctan \frac{c}{2Y}, \tag{1.3}$$
$$D_0 = \sum_j \nu_j F_{c_j}(Y) - \Theta Y - m, \tag{1.4}$$
$$G_0 = m + \left(\sum_j \nu_j c_j \right) \log 2, \tag{1.5}$$
$$D = D_0 + \gamma, \qquad G = G_0 - \gamma. \tag{1.6}$$

Theorem 1.1 (Odd-power bound)**.** *Let $k \geq 3$ be odd. If $G \geq 0$ and $\Delta_k = D - (k-1)G > 0$, then*
$$\boxed{\mu(\pi^k) \leq E_k := \frac{m(D+G)}{D-(k-1)G} = \frac{m(D_0+G_0)}{D_0-(k-1)G_0+k\gamma}.} \tag{1.7}$$

The proof also works for every integer $k \geq 2$. We emphasize odd powers because they are the motivating targets; we do not compare these bounds with specialized results for even powers.

Corollary 1.2 (A twelve-layer family)**.** *Use $L = 69$, $t_j = j$ for $0 \leq j \leq 11$, and*

$$\nu_0 = 6s, \qquad \nu_1 = \cdots = \nu_{11} = s, \qquad m = 17s - 1.$$

The following choices satisfy the hypotheses:

k	s	m	*Certified strict upper bound*
3	*11*	*186*	1203.973
5	*259*	4 402	42 461
7	5 871	99 806	1 272 103
9	131 354	2 233 017	35 398 937

For every other fixed odd k, some sufficiently large integer s also satisfies the hypotheses.

The last assertion is an existence statement within an explicit family, not a uniform small bound as $k \to \infty$. Section 6 explains the growth.

1.5 A guide to the evidence

The algebraic and analytic lemmas prove Theorem 1.1. Ball-arithmetic certificates prove the decimal comparisons once the lemmas hold. Exact finite examples check identities and implementation, but do not prove asymptotic rates. Floating-point searches select parameters; they neither certify rounding nor prove global optimality. The appendices retain unsuccessful experiments because they explain the arithmetic difficulties that remain.

2 The auxiliary polynomials and their determinant

2.1 Intervals, poles, and normalization

Fix the parameters of Theorem 1.1. Let n run through positive multiples of L. List all $M = m + 1$ intervals, including repetitions, in nested order:

$$I_a = [\alpha_a, \beta_a], \quad 0 \leq a \leq m, \qquad 0 = \alpha_0 \leq \cdots \leq \alpha_m < n/2, \quad \beta_a = n - \alpha_a.$$

Thus $I_m \subseteq \cdots \subseteq I_0 = [0, n]$. Write $\ell_a = \beta_a - \alpha_a$. The grouped description uses endpoints $t_j n/L, (L - t_j)n/L$, with multiplicity ν_j.

Set

$$\begin{aligned} Q_n(z) &= \prod_{a=0}^{m} \prod_{r=\alpha_a}^{\beta_a} (z + r), \\ N_n &= \sum_{a=0}^{m} \ell_a, \qquad d_n = \operatorname{lcm}(1, \ldots, n), \\ C_n &= m!\, d_n^m \prod_{a=0}^{m} \ell_a!. \end{aligned} \tag{2.1}$$

The degree of Q_n is $N_n + M$. *The interval factorials are in the numerator of C_n.*

Let Γ positively surround the poles $0, -1, \ldots, -n$. On a local logarithm branch define

$$R_{h,n}(x) = \frac{C_n}{2\pi \mathrm{i}} \oint_\Gamma \frac{z^h x^{z+n}}{Q_n(z)} \, \mathrm{d}z, \qquad 0 \le h \le m. \tag{2.2}$$

Write

$$\frac{z^h}{Q_n(z)} = \sum_{\ell=0}^{n} \sum_{r=1}^{\mu_\ell} \frac{b_{h,\ell,r}}{(z+\ell)^r}, \quad \mu_\ell = \#\{a : \ell \in I_a\}.$$

Residues give

$$R_{h,n}(x) = \sum_{j=0}^{m} A_{h,j,n}(x)(\log x)^j, \tag{2.3}$$

$$A_{h,j,n}(x) = \frac{C_n}{j!} \sum_{\ell : \mu_\ell \ge j+1} b_{h,\ell,j+1} x^{n-\ell}. \tag{2.4}$$

2.2 Why all coefficients are integers

Lemma 2.1 (Integrality)**.** *For all h, j, one has $A_{h,j,n}(x) \in \mathbb{Z}[x]$.*

Proof. Absorb the interval factorials first:

$$H_a(z) = \frac{\ell_a!}{\prod_{r=\alpha_a}^{\beta_a}(z+r)}.$$

If $\ell \in I_a$, expansion at $z = -\ell + w$ gives

$$wH_a(-\ell+w) = (-1)^{\ell-\alpha_a} \binom{\ell_a}{\ell-\alpha_a} \prod_{\substack{r \in I_a \\ r \ne \ell}} \left(1 + \frac{w}{r-\ell}\right)^{-1}. \tag{2.5}$$

The coefficient of w^u has denominator dividing d_n^u: every distance is a nonzero integer of absolute value at most n.

If $\ell \notin I_a$, use

$$H_a(z) = \sum_{r=\alpha_a}^{\beta_a} \frac{(-1)^{r-\alpha_a}\binom{\ell_a}{r-\alpha_a}}{z+r}. \tag{2.6}$$

Now the coefficient of w^u has denominator dividing d_n^{u+1}. The extra factor matters: a regular factor need not have an integral constant term.

There are μ_ℓ pole factors and $M - \mu_\ell$ regular factors. The coefficient of $(z+\ell)^{-j-1}$ requires total local order $\mu_\ell - 1 - j$. The polynomial $(-\ell+w)^h$ has integer coefficients and can only reduce the order required from the other factors. Thus the necessary exponent of d_n is at most

$$(\mu_\ell - 1 - j) + (M - \mu_\ell) = m - j.$$

Multiplication by $m! d_n^m / j!$ clears the denominators. □

The cancellation of μ_ℓ is why a variable pole multiplicity does not require a larger universal power than d_n^m.

2.3 Supports and vanishing orders

Nestedness implies

$$\mu_\ell \geq j+1 \quad \Longleftrightarrow \quad \ell \in [\alpha_j, \beta_j].$$

Hence $A_{h,j,n}$ is divisible by $x^{n-\beta_j}$, and the remaining degree is at most ℓ_j.

Lemma 2.2 (Vanishing at 1)**.** *Every $R_{h,n}$ has a zero of order at least N_n at $x = 1$.*

Proof. Put $w = \log x$. The coefficient of w^r in the integral of $z^h e^{zw}/Q_n(z)$ is zero whenever $h + r \leq \deg Q_n - 2$, by the residue at infinity. The first possible order is therefore

$$\deg Q_n - h - 1 = N_n + M - h - 1 \geq N_n.$$

Multiplication by x^n does not change this order. Also $\log x = (x-1) + O((x-1)^2)$. □

2.4 A nonzero determinant, including its leading term

Proposition 2.3 (Exact determinant)**.** *There exists $c_n \neq 0$ such that*

$$\det(A_{h,j,n}(x))_{0\leq h,j\leq m} = c_n x^{\sum_j (n-\beta_j)} (x-1)^{N_n}. \tag{2.7}$$

In particular $A_n(\mathrm{i})$ is invertible.

Proof. The supports give the power of x. After it is removed, the determinant has degree at most N_n. Replace column zero by the sum of column j multiplied by $(\log x)^j$, for $0 \leq j \leq m$. Its determinant is unchanged; the new column consists of the $R_{h,n}(x)$. Lemma 2.2 supplies the factor $(x-1)^{N_n}$.

To prove nonvanishing, take a block of repeated left endpoints

$$\alpha_u = \cdots = \alpha_v = \alpha, \qquad r = v - u + 1.$$

The pole at $-\alpha$ has multiplicity $v+1$. Write

$$\frac{1}{Q_n(-\alpha + w)} = w^{-(v+1)} B_\alpha(w), \qquad B_\alpha(0) \neq 0.$$

For $u \leq j \leq v$, the coefficient of $x^{n-\alpha}$ in column j is

$$\frac{C_n}{j!}[w^{v-j}]((-\alpha+w)^h B_\alpha(w)).$$

The orders $v - j$ are $0, \ldots, r-1$, in reverse order. Multiplication by B_α gives an invertible triangular change on these functionals.

In all blocks together, the leading-coefficient matrix reduces to the divided-derivative evaluations of $1, z, \ldots, z^m$ at distinct points $-\alpha$. This confluent Vandermonde matrix is invertible: a polynomial in its kernel would have zeros of total multiplicity $M = m+1$, exceeding its degree. Thus the leading determinant is nonzero, proving the identity. □

High vanishing order and a degree bound alone would also allow the zero determinant. The leading-term argument rules out that possibility.

3 Prime divisors and weighted zero arcs

3.1 A local valuation inequality

For a rational number $u \neq 0$, let $\mathrm{ord}_p(u)$ be its usual p-adic valuation, and put $\mathrm{ord}_p(0) = +\infty$. We only use primes $\sqrt{n} < p \leq n$, so a nonzero distance of absolute value at most n has valuation at most one.

Lemma 3.1 (A coefficientwise divisor)**.** *For such a prime and an integer $0 \leq \ell \leq n$, put*

$$w_a(p,\ell) = \left\lfloor \frac{\ell_a}{p} \right\rfloor - \left\lfloor \frac{\ell - \alpha_a}{p} \right\rfloor - \left\lfloor \frac{\beta_a - \ell}{p} \right\rfloor.$$

Each w_a is 0 or 1. Every coefficient of $x^{n-\ell}$ in $A_{h,j,n}$ has valuation at least

$$\sum_{a=0}^{m} w_a(p,\ell) + j. \tag{3.1}$$

Proof. The assertion $w_a \in \{0,1\}$ follows from $\lfloor u+v \rfloor - \lfloor u \rfloor - \lfloor v \rfloor \in \{0,1\}$, with $u = (\ell - \alpha_a)/p$ and $v = (\beta_a - \ell)/p$.

For a pole factor, the constant in (2.5) is a binomial coefficient of valuation exactly w_a, by the factorial valuation formula and $p^2 > n$. Its coefficient of order u has valuation at least $w_a - u$.

For a regular factor the constant term has valuation $w_a - 1$. For example, if $\ell < \alpha_a$ and $s = \alpha_a - \ell$, it equals

$$H_a(-\ell) = \frac{\ell_a!(s-1)!}{(s+\ell_a)!}.$$

Use $\lfloor (s-1)/p \rfloor = -\lfloor -s/p \rfloor - 1$; the other side of the interval is identical up to sign. Expansion of the reciprocal linear factors loses at most one valuation per order, so order u has valuation at least $w_a - 1 - u$.

The needed total order is at most $\mu_\ell - 1 - j$, and there are $M - \mu_\ell$ regular factors. Since $\mathrm{ord}_p(d_n) = 1$, the lower bound after multiplication by d_n^m is

$$m + \sum_a w_a - (M - \mu_\ell) - (\mu_\ell - 1 - j) = \sum_a w_a + j.$$

The factor $m!/j!$ and coefficients of $(-\ell + w)^h$ are integers and cannot lower this bound. Sums of terms preserve the lower bound. □

This estimate is stronger than needed. We discard the extra j and minimize over the pole location in order to obtain a divisor common to all entries of the matrix.

3.2 Replacing pole locations by a circle

Write $n = LN_0$. For the grouped intervals define

$$W_j(z,y) = \lfloor (L - 2t_j)z \rfloor - \lfloor y - t_j z \rfloor - \lfloor (L - t_j)z - y \rfloor. \tag{3.2}$$

It is periodic in z and y with period one and takes values in $\{0,1\}$. At $z = N_0/p,\ y = \ell/p$, it is precisely the local w_a for an interval of type j.

For generic z, the set on which $W_j(z,y)=0$ is the closed counterclockwise arc from $\{t_j z\}$ to $\{(L-t_j)z\}$. Its length is $\{(L-2t_j)z\}$, where braces denote fractional part. Thus

$$f(z)=\min_{0\le y<1}\sum_j \nu_j W_j(z,y)=M-\max_y\{\text{total weight of zero arcs containing } y\}. \tag{3.3}$$

The weights are the interval multiplicities ν_j.

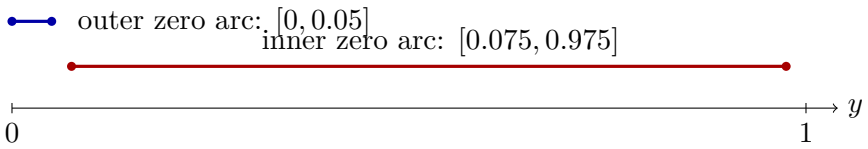


Figure 1: For $L=14,\ t_0=0,\ t_1=1,\ z=3/40$, the zero arcs are disjoint. With equal weights r, their common-complement cost is $f(z)=r$: at least one type contributes r at every y.

All endpoints are affine integer multiples of z, modulo one. Their cyclic order changes only at the following finite rational set. Let

$$\mathcal{S}=\{0\}\cup\{t_j, L-t_j : 0\le j<J\},\qquad \mathcal{D}=\{|u-v| : u,v\in\mathcal{S},\ u\ne v\}.$$

Take all fractions $a/d\in[0,1]$ with $d\in\mathcal{D}$, remove duplicates, and list them as

$$0=u_0<u_1<\cdots<u_R=1.$$

On each open interval (u_r,u_{r+1}), the integer $f(z)$ is constant. At a fixed rational midpoint z, it can be calculated exactly by checking midpoints between consecutive arc endpoints. Off this exceptional set no distinct endpoints coincide, so an endpoint cannot have a smaller value than both adjacent open arcs.

To avoid endpoint conventions entirely, define $f=0$ at every breakpoint and extend it periodically. This only weakens the local divisor estimate and does not change any integral below.

3.3 The common divisor and its exact exponential rate

Define

$$\Pi_n=\prod_{\sqrt{n}<p\le n} p^{f(N_0/p)}. \tag{3.4}$$

By Lemma 3.1, Π_n divides every integer coefficient of every $A_{h,j,n}(x)$. It is a rational integer; we do not need factorization into Gaussian primes.

Proposition 3.2 (Digamma formula)**.** *The limit* $\gamma=\lim n^{-1}\log\Pi_n$ *exists and is*

$$\boxed{\gamma=\frac{1}{L}\sum_{\substack{0\le r<R\\ u_r>0}} f\!\left(\frac{u_r+u_{r+1}}{2}\right)\big(\psi(u_{r+1})-\psi(u_r)\big),} \tag{3.5}$$

where $\psi=\Gamma'/\Gamma$.

Proof. For all sufficiently small $z>0$, none of the endpoints wraps around the circle and the zero arcs are nested about $Lz/2$. They have a common point, so $f(z)=0$. Thus the apparent singularity at $u_0=0$ contributes nothing.

For an interval $(a,b) \subset (0,1)$ on which $f = v$, periodicity selects primes satisfying

$$N_0/(r+b) < p < N_0/(r+a), \qquad r = 0, 1, \ldots.$$

For each fixed r, the prime number theorem $\vartheta(X) = \sum_{p \le X} \log p = X + o(X)$ makes their contribution to $n^{-1} \log \Pi_n$ tend to

$$\frac{v}{L}\left(\frac{1}{r+a} - \frac{1}{r+b}\right).$$

The sum over $r \ge T$ is bounded, uniformly in large n, by a constant times $M/(LT)$, using the elementary Chebyshev bound $\vartheta(X) = O(X)$. Primes below $\sqrt{n}$ contribute $o(n)$. These observations justify passage from finitely many periods to all periods.

Breakpoint exclusions also cost only $O(\log n)$. Indeed, $\{N_0/p\} = a/d$ implies that p divides dN_0; the denominators d belong to a fixed finite set. Finally,

$$\sum_{r=0}^{\infty}\left(\frac{1}{r+a} - \frac{1}{r+b}\right) = \psi(b) - \psi(a).$$

Summing over the finite partition proves the formula. □

The formula is a lower bound on the content available for division, not a formula for the entire actual gcd of a row after evaluation at $x = \mathrm{i}$. Additional cancellations at i could give more divisibility.

3.4 A transparent two-layer special case

For $t_0 = 0, t_1 = 1$ with weights $r, M - r$, put

$$a_\ell = \max\left(\frac{\ell}{L}, \frac{\ell - 1}{L-2}\right), \qquad b_\ell = \frac{\ell}{L-1}, \quad 1 \le \ell \le L-2.$$

On these intervals the two zero arcs are disjoint, and $f = \min(r, M-r)$. They are precisely the disjointness intervals away from endpoints. Hence

$$\gamma = \frac{\min(r, M-r)}{L} \sum_{\ell=1}^{L-2} (\psi(b_\ell) - \psi(a_\ell)). \tag{3.6}$$

This is a useful small test of the general partition algorithm. For many layers, the weighted maximum-overlap calculation is more effective than checking pairwise disjointness alone.

4 Uniform analytic estimates

4.1 A coefficient bound valid on a whole disk

Lemma 4.1 (Height and divided difference)**.** *Uniformly for $0 \le h, j \le m$,*

$$|A_{h,j,n}(\mathrm{i})| \le \exp(G_0 n + o(n)).$$

After division by Π_n, put

$$U_{h,n}(Y) = \sum_{j=0}^{m} \frac{A_{h,j,n}(\mathrm{i})}{\Pi_n} Y^j \in \mathbb{Z}[\mathrm{i}][Y],$$

and define the polynomial

$$T_{h,n}(Y) = \sum_{j=1}^{m} \frac{A_{h,j,n}(\mathrm{i})}{\Pi_n} \sum_{a=0}^{j-1} \lambda^{j-1-a} Y^a.$$

For $|Y| \le 2$, uniformly in h, Y,

$$|U_{h,n}(Y)|,\ |T_{h,n}(Y)| \le \exp(Gn + o(n)), \tag{4.1}$$

and

$$U_{h,n}(\lambda) - U_{h,n}(Y) = (\lambda - Y) T_{h,n}(Y). \tag{4.2}$$

Proof. In (2.5), the constant has modulus at most 2^{ℓ_a}. For a regular factor the constant has modulus at most one, since it is, up to sign, $\ell_a!(s-1)!/(s+\ell_a)!$ for an integer $s \ge 1$. For either type, the remaining reciprocal linear factors have nonzero integral distances. A coefficient of fixed order at most m is bounded by the corresponding coefficient of $(1-w)^{-(N_n+M)}$, and is therefore polynomial in n. The factors $(-\ell + w)^h$, and the sum over ℓ, introduce only more fixed powers of n. Thus

$$|A_{h,j,n}(\mathrm{i})| \le K\, d_n^m n^B 2^{\sum_a \ell_a}.$$

Since $\log d_n = n + o(n)$ and $\sum_a \ell_a = n \sum_j \nu_j c_j$, this has rate G_0. Division by Π_n subtracts γ. The degree m and the bounds $|Y| \le 2$, $|\lambda| < 2$ are fixed, so evaluation and the displayed finite difference introduce only constant factors. The identity follows from $\lambda^j - Y^j = (\lambda - Y) \sum_{a=0}^{j-1} \lambda^{j-1-a} Y^a$. □

This is a uniform estimate on a disk, not just an asymptotic value at a single point. Uniformity will permit a rational approximation to select its own nearby algebraic point $Y = y$.

4.2 A global horizontal-contour bound

The main integral estimate can be proved without leaving its tails implicit. First observe that the solution of (1.2) is unique: its left side decreases continuously from $M\pi$ to zero. Also

$$Y > \frac{1}{2}. \tag{4.3}$$

Indeed the outer type has $c_0 = 1, \nu_0 \ge 1$, and $M \ge 2$; at $Y = 1/2$ that type alone contributes at least $\pi/2$, and the remaining multiplicity makes the inequality strict.

Lemma 4.2 (A uniform tail inequality)**.** *For $Y > 1/2$, put*

$$B_Y = \frac{(Y^2 + 1/4)^2}{Y^2 - 1/4}.$$

If $d = \deg Q_n = N_n + M$, then for every real t,

$$\frac{|Q_n(-n/2 + nt + \mathrm{i}nY)|}{|Q_n(-n/2 + \mathrm{i}nY)|} \ge (1 + t^2/B_Y)^{d/2}. \tag{4.4}$$

The same inequality holds on the lower horizontal line.

Proof. The offsets $u = r/n - 1/2$ in each interval occur in pairs $u, -u$, with $|u| \le 1/2$. For a pair, the square of the modulus ratio is

$$1 + \frac{2(Y^2 - u^2)t^2 + t^4}{(Y^2 + u^2)^2}.$$

It is at least $(1+t^2/B_Y)^2$: the coefficient of t^2 is at least $2/B_Y$, and that of t^4 is at least $1/B_Y^2$. An unpaired central factor has ratio $\sqrt{1+t^2/Y^2} \geq \sqrt{1+t^2/B_Y}$. Multiplication proves the claim, with repetitions included. □

Open the contour in (2.2) to the lines $\operatorname{Im} z = \pm nY$. For each fixed n the vertical ends vanish as their real parts tend to infinity: the rational factor decays faster than $1/|z|$. On these horizontal lines, $|\mathrm{i}^{z+n}| \leq e^{\Theta nY}$. Lemma 4.2 therefore gives

$$|R_{h,n}(\mathrm{i})| \leq K_{h,Y} n^{h+1} e^{\Theta nY} \frac{C_n}{|Q_n(-n/2+\mathrm{i}nY)|}. \tag{4.5}$$

To justify a constant independent of n, the remaining integral is bounded by

$$\int_{\mathbb{R}} \frac{(|t|+Y+1/2)^h}{(1+t^2/B_Y)^{d/2}}\,\mathrm{d}t.$$

For sufficiently large n, replace the denominator exponent by the fixed value $h+2$; the resulting integral is finite. This controls the entire line, including both infinite tails.

4.3 Evaluating the central exponential rate

For one interval of length cn, Stirling's formula and a Riemann sum on a smooth integrand give

$$\log((cn)!) = cn\log n + cn\log c - cn + O(\log n),$$

$$\sum_{r=\alpha}^{\beta} \log|r - n/2 + \mathrm{i}nY| = (cn+1)\log n + n\int_{-c/2}^{c/2} \log|u+\mathrm{i}Y|\,\mathrm{d}u + O(1).$$

The extra endpoint in the product changes only $O(\log n)$. Direct integration shows

$$\int_{-c/2}^{c/2} \log|u+\mathrm{i}Y|\,\mathrm{d}u = \frac{c}{2}\log(Y^2+c^2/4) - c + 2Y\arctan\frac{c}{2Y}.$$

Subtracting the product logarithm from the factorial logarithm thus gives $-nF_c(Y)+O(\log n)$. Equation (4.5) yields

$$|R_{h,n}(\mathrm{i})| \leq \exp\left\{-n\left(\sum_j \nu_j F_{c_j}(Y) - \Theta Y - m\right) + o(n)\right\}.$$

Proposition 4.3 (Small remainders)**.** *After division by* Π_n*, uniformly in* $0 \leq h \leq m$*,*

$$|U_{h,n}(\lambda)| \leq \exp(-Dn + o(n)). \tag{4.6}$$

Finally,

$$\frac{\mathrm{d}}{\mathrm{d}Y}F_c(Y) = 2\arctan\frac{c}{2Y}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}Y^2}F_c(Y) = -\frac{c}{Y^2+c^2/4}.$$

Hence (1.2) is the unique global maximizer of the central decay bound. This optimization needs only a one-dimensional monotone root, not a claimed exact asymptotic formula for an oscillating complex sum.

5 From logarithmic polynomials to odd powers

5.1 Why a resultant is the right arithmetic object

The Gaussian integers $\mathbb{Z}[\mathrm{i}]$ form a unique factorization domain. Every nonzero Gaussian integer has complex modulus at least one. If f is a primitive polynomial over this ring and divides a polynomial over $\mathbb{Q}(\mathrm{i})$, Gauss's lemma makes that divisibility valid over $\mathbb{Z}[\mathrm{i}]$ as well.

For polynomials f, U, the resultant vanishes exactly when they have a common root. If f has leading coefficient a and roots $y_1, \dots, y_d$, then

$$\operatorname{Res}(f, U) = a^{\deg U} \prod_{\ell=1}^{d} U(y_\ell).$$

Using a *minimal* polynomial, rather than the whole binomial that first defines y, guarantees the nonzero product needed below.

Lemma 5.1 (Uniform algebraic-value bound)**.** *Let $k \geq 2$, let $p/q > 0$ be reduced, and suppose $|p/q - \pi^k| < 1$. Put*

$$\theta = (p/q)^{1/k} > 0, \qquad y = \mathrm{i}\theta/2.$$

For every n, some h has $U_{h,n}(y) \neq 0$. For each $\eta > 0$ and all sufficiently large n, this h satisfies

$$|U_{h,n}(y)| \geq (2^k q)^{-m} \exp(-(k-1)(G+\eta)n). \tag{5.1}$$

The threshold is independent of p, q.

Proof. Dividing the matrix by Π_n preserves its invertibility. It cannot send the nonzero vector $(1, y, \dots, y^m)^{\mathsf{T}}$ to zero, so some row U is nonzero at y.

Let $f \in \mathbb{Z}[\mathrm{i}][Y]$ be a primitive minimal polynomial of y over $K = \mathbb{Q}(\mathrm{i})$, of degree $d \leq k$, and let $a \neq 0$ be its leading coefficient. It divides

$$(2^k q)Y^k - \mathrm{i}^k p$$

over $\mathbb{Z}[\mathrm{i}]$. Thus $a \mid 2^k q$, in particular $|a| \leq 2^k q$. Every root of f belongs to $\{y\zeta : \zeta^k = 1\}$. The error assumption implies $|y| < 2$, so all these roots lie in the disk used in Lemma 4.1.

Irreducibility of f and $U(y) \neq 0$ give $\operatorname{Res}(f, U) \neq 0$. Its Gaussian-integrality yields

$$1 \leq |a|^{\deg U} |U(y)| \prod_{\ell=2}^{d} |U(y_\ell)|.$$

Use $\deg U \leq m$, $d \leq k$, and the uniform height bound. Since $G + \eta > 0$, replacing $d - 1$ by $k - 1$ only weakens the result and gives (5.1). □

No assumption of irreducibility of $Y^k - \mathrm{i}^k p/(2^k q)$ has been made. If p/q is a rational power, or a binomial splits over K, the true degree d is simply smaller. In particular, the argument covers every denominator and not just a generic subsequence of approximations.

5.2 The quantitative comparison

Proof of Theorem 1.1. Write $\Delta_k = D - (k-1)G > 0$. Choose $\eta > 0$ small enough that $\Delta_k - k\eta > 0$. All estimates can be made simultaneous: for $n \geq n_0$, every row and every $|Y| \leq 2$ satisfy

$$|U_{h,n}(\lambda)| \leq e^{-(D-\eta)n}, \quad |U_{h,n}(Y)|,\ |T_{h,n}(Y)| \leq e^{(G+\eta)n}.$$

Here n was required to be a multiple of L. Accordingly choose

$$n = L\left\lceil \frac{m\log(2^k q) + \log 2}{L(\Delta_k - k\eta)} \right\rceil. \tag{5.2}$$

For sufficiently large q, this is at least n_0. The extra rounding is at most L, a fixed constant.

Select the row in Lemma 5.1. The ratio of the small remainder to its nonzero value at y is at most

$$(2^k q)^m e^{-(\Delta_k - k\eta)n} \le \tfrac{1}{2}.$$

The divided-difference identity consequently gives

$$|\lambda - y| \ge \tfrac{1}{2}(2^k q)^{-m} e^{-k(G+\eta)n}.$$

If the divided difference were zero, that same identity would contradict the preceding factor $1/2$; hence no division by zero is hidden here.

Substitution of (5.2) yields

$$|\lambda - y| \ge c_{k,m,\eta} q^{-m(D+G)/(\Delta_k - k\eta)}.$$

Indeed $\Delta_k - k\eta + k(G+\eta) = D + G$, which explains the numerator in the final exponent.

Finally,

$$\left|\pi^k - \frac{p}{q}\right| = |\pi - \theta| \sum_{a=0}^{k-1} \pi^{k-1-a}\theta^a \ge 2\pi^{k-1}|\lambda - y|.$$

Fractions outside $|p/q - \pi^k| < 1$, or with $p/q \le 0$, need no special treatment. Letting η tend to zero proves the irrationality-exponent assertion. □

5.3 Why a common divisor helps twice

Before division, the rates are D_0, G_0. Dividing all coefficients and remainders by the same Π_n changes them to

$$D = D_0 + \gamma, \qquad G = G_0 - \gamma.$$

Thus the numerator $D + G$ is unchanged, while the useful gap grows by $k\gamma$:

$$D - (k-1)G = D_0 - (k-1)G_0 + k\gamma.$$

This is why a divisor must be applied consistently to the remainder, the coefficients, and the divided difference. Reducing only one height in the final formula would give an incorrect improvement.

6 Concrete parameters and every fixed odd power

6.1 Recovering the balanced benchmark

If all M intervals equal $[0, n]$, the arc minimum used here is zero, so $\gamma = 0$ for this particular divisor. This does not prove that the entire actual gcd has rate zero. The root equation becomes

$$Y = \tfrac{1}{2}\cot\frac{\pi}{4M}.$$

Substitution gives

$$D_0 = -M\log\left(2\sin\frac{\pi}{4M}\right) - m, \qquad G_0 = m + M\log 2.$$

With $m = 289$, the resulting exponent is $2028.145382310168\ldots$. This is a useful check of the normalization and contour calculation.

6.2 Two, seven, and twelve layers

Three configurations illustrate the improvement:

Configuration	m	L	Offsets and multiplicities
Two layers	255	14	$t = (0, 1),\ \nu = (128, 128)$
Seven layers	199	39	$t = (0, \dots, 6),\ \nu = (80, 20, \dots, 20)$
Twelve layers	186	69	$t = (0, \dots, 11),\ \nu = (66, 11, \dots, 11)$

Their rates for $k = 3$, rounded here only for readability, are:

Configuration	D_0	G_0	γ	E_3
Two layers	955.116623	419.770987	27.777682	1762.608232
Seven layers	664.537005	322.700112	44.701426	1282.033606
Twelve layers	605.861295	301.032295	45.436001	1203.972439

The double-layer result follows also from (3.6). For the twelve-layer case the corrected rates are

$$\begin{aligned} D &= 651.297296270676769392603641659 9\dots, \\ G &= 255.596293834458504937833052789 8\dots, \\ D - 2G &= 140.104708601759759516937536080 1\dots. \end{aligned}$$

The strict conclusion is $E_3 < 1203.973$, not a conclusion inferred from the rounded entries of the table.

These examples arose from bounded searches and a limited multiplicity refinement. They are not the solution of a globally optimized variational problem. In particular, more layers are not known to be necessary or optimal.

6.3 An explicit family for all odd k

For the twelve-layer shape let

$$v_0 = 6, \quad v_j = 1\ (1 \le j \le 11), \quad c_j = 1 - \frac{2j}{69}, \quad \nu_j = s v_j.$$

Then

$$M = 17s, \quad m = 17s - 1, \quad S = \sum_j v_j c_j = \frac{347}{23}.$$

The arc partition is unchanged when all weights are multiplied by s, so

$$\gamma = s\gamma_*, \qquad \gamma_* = 4.130545573141366136266461634 1\dots.$$

Introduce the fixed constants

$$g_* = 17 + S\log 2 - \gamma_*, \qquad g_* = 23.326935803132591357984822980 8\dots, \tag{6.1}$$

$$A_* = \sum_j v_j c_j \log c_j,$$

$$d_* = S\log(S/\Theta) - A_* - 17 + \gamma_*. \tag{6.2}$$

Since $f \le 17$ for the base weights, the prime number theorem gives $\gamma_* \le 17$. Thus $g_* \ge S\log 2 > 1$, proving $G = sg_* - 1 > 0$ without requiring a decimal estimate. The certificate more precisely verifies $g_* > 23$.

Proposition 6.1 (A root-free admissibility recipe)**.** *For any fixed integer $k \geq 3$, let*

$$s_k = \max\left\{1,\ \left\lceil \exp\left(\frac{(k-1)g_* - d_*}{S} + 1\right)\right\rceil\right\}. \tag{6.3}$$

This choice satisfies $D - (k-1)G > 0$. More explicitly, put

$$\underline{D}_s = s(S\log s + d_*) + 1 - \frac{\Theta^2}{12sS}, \qquad G_s = sg_* - 1.$$

Whenever $\underline{D}_s > (k-1)G_s$, the following bound needs no numerical root Y:

$$\mu(\pi^k) \leq (17s-1)\frac{\underline{D}_s + G_s}{\underline{D}_s - (k-1)G_s}. \tag{6.4}$$

In particular this is valid at $s = s_k$.

Proof. Evaluate the decay expression at the trial height $Y_0 = sS/\Theta$; the optimized D is at least this trial value. For every $x \geq 0$, $\arctan x \geq x - x^3/3$ and $\log(1+x^2) \geq 0$. Consequently

$$F_c(Y_0) \geq c\log(Y_0/c) + c - \frac{c^3}{12Y_0^2}.$$

Since $0 < c_j \leq 1$, one has $\sum_j v_j c_j^3 \leq S$. The terms $sS - \Theta Y_0$ cancel, giving

$$D \geq s(S\log s + d_*) + 1 - \frac{\Theta^2}{12sS} = \underline{D}_s.$$

At $s = s_k$,

$$\underline{D}_s - (k-1)G_s \geq sS + k - \frac{\Theta^2}{12sS} > 0.$$

Here $S = 347/23$, $s \geq 1$, and $\Theta < 2$, so the last positivity is immediate.

For $G_s \geq 0$ the function $m(D+G_s)/(D-(k-1)G_s)$ decreases as D increases. Replacing D by the lower bound $\underline{D}_s$ therefore gives (6.4). □

This proposition supplies a definite upper-bound recipe for *every fixed odd power*, in addition to the optimized examples in Corollary 1.2. It does not require an unbounded parameter search to establish admissibility.

6.4 Why the higher-power constants still grow rapidly

Expanding the root equation for large s gives

$$Y = \frac{sS}{\Theta} + O(s^{-1}), \quad D = s(S\log s + d_*) + 1 + O(s^{-1}), \quad G = sg_* - 1.$$

Thus the threshold for s grows exponentially with k. The explicit choice (6.3) yields

$$E_k = O(k\rho^k), \qquad \rho = \exp(g_*/S) = 4.69343992780744986545\ldots. \tag{6.5}$$

To justify the order estimate directly, that choice has $s_k = O(\rho^k)$, a gap at least $s_k S + O(1)$, and numerator $(17s_k - 1)(\underline{D}_{s_k} + G_{s_k}) = O(ks_k^2)$. All implied constants are fixed by this twelve-layer shape. Equation (6.5) describes the construction as k varies; it must not be confused with the separate limit $n \to \infty$ used to prove each fixed-k bound.

For orientation, the certified optimized values before upward rounding are

k	E_k
3	1203.97243920634698117...
5	42460.70466848845471678...
7	1272102.82718073869325320...
9	35398936.33162709355243184...

The number s was optimized only within one fixed shape.

6.5 What would be needed to reach 10–100?

In the present transfer formula, $G \geq 0$ implies

$$E_k = m\left(1 + \frac{kG}{D-(k-1)G}\right) \geq m.$$

The best current example has $m = 186$. Hence this particular parameter choice cannot possibly give $E_3 < 100$. This is not an impossibility theorem for all nested shapes.

A more substantial change would keep the *target polynomial degree* fixed at three. If

$$|P(\lambda)| \geq c_\varepsilon \mathrm{H}(P)^{-\nu-\varepsilon} \quad (P \in \mathbb{Z}[\mathrm{i}][Y],\ \deg P \leq 3,\ P \neq 0),$$

then taking $P(Y) = 8qY^3 + \mathrm{i}p$ gives

$$|P(\lambda)| = q|\pi^3 - p/q|, \qquad \mu(\pi^3) \leq \nu + 1.$$

Only this sparse binomial family is actually needed. The thresholds $\nu \leq 99$ and $\nu \leq 9$ correspond to upper bounds 100 and 10. Type II Hermite–Padé and multiple orthogonal polynomial constructions are plausible research tools; examples of bounded-degree logarithmic approximation appear in Marcovecchio [3]. Their numerical constants at other arguments cannot be transplanted to $\log \mathrm{i}$ without checking the complex endpoint and its arithmetic.

7 How the constants are computed and certified

7.1 An exact finite combinatorial calculation

The common divisor does not require numerical integration over a circle. The calculation is finite:

1. Form the integer slopes $0, t_j, L - t_j$ and all nonzero differences.
2. Form and sort the rational critical points $a/d \in [0, 1]$.
3. In each open critical cell choose its rational midpoint z. Sort all fractional endpoints $\{t_j z\}, \{(L - t_j)z\}$.
4. Evaluate the integer vector $W(z, y)$ at each rational midpoint between consecutive endpoints. Minimize its weighted sum.
5. Apply the finite digamma sum (3.5). Adjacent cells with the same minimum may be merged.

Only the final special-function evaluations use real arithmetic. All endpoint orderings and minima use exact fractions and integers.

For the base weights $(6, 1, \ldots, 1)$ at $L = 69$, the first merged nonzero cells are:

Left endpoint	Right endpoint	f
1/69	1/67	6
1/67	1/65	7
1/65	1/62	8
1/62	1/60	7
1/60	1/58	6
1/58	1/56	5
1/56	1/54	4
1/54	1/52	3

These are only the beginning of the partition. Endpoint wraparound and changing overlaps make a full computation preferable to extrapolating a pattern from the first few cells. The complete merged list is supplied in the certificate data.

7.2 Ball arithmetic, root signs, and strict rounding

The standalone program `code/certify.py` requires Python and `python-flint`. It uses Arb [4] at 256-bit precision. A ball encloses the exact real value by a midpoint and a rigorously propagated error radius.

The saddle equation is strictly decreasing. At $Y = 1/2$ its left side minus Θ is positive, and at $Y = 2M$ it is negative. The program performs 210 rational bisections, accepting each sign only if the entire corresponding ball has that sign. It then evaluates $D_0, G_0, \gamma, D, G, \Delta_k, E_k$ on the resulting interval and asserts both $\Delta_k > 0$ and the advertised strict upper bound.

For example, in the twelve-layer cubic case it encloses

$$E_3 = 1203.97243920634698117089090450914656\ldots$$

with an error radius below $6 \cdot 10^{-59}$. The assertion $E_3 < 1203.973$ has a wide safety margin relative to this radius. The certificate does not assume that a floating-point optimizer located a true minimum.

The same program checks the arc bound independently against all integer pole positions for $n = 345$: 52 nonexceptional primes are checked, and 39 have a positive guaranteed divisor exponent. This finite check tests the combinatorial implementation; the proof for arbitrary n is Lemma 3.1.

7.3 Reproduction and the role of each file

Run the following from the project directory:

```
python code/certify.py
python code/finite_checks.py
xelatex -interaction=nonstopmode -halt-on-error main.tex
xelatex -interaction=nonstopmode -halt-on-error main.tex
```

The first command regenerates `data/certificates.json`. The second requires SymPy and checks the small exact examples in Appendix A. Optional experiment programs require the additional packages stated in the README. They are not prerequisites for the main proof.

The certificate records the parameters, merged nonzero arc cells, root enclosure, all rates, the gap, and the strict comparisons. Its purpose is to make the numerical part reproducible. The prime number theorem and the asymptotic inequalities are mathematical inputs, not claims checked by a finite program.

7.4 Parameter order and effectivity

For a chosen odd k, first fix L, t_j, ν_j, or fix the scale s in the twelve-layer family. Then fix an exponent slack ε, and choose the small η in Section 5. Only after these choices does n tend to infinity, and only then is n selected as a function of the approximation denominator q.

This manuscript gives explicit upper bounds on the irrationality *exponent*. It does not tabulate an explicit numerical $q_0(\varepsilon)$ or the constants in (1.1). Those would require explicit versions of the prime-counting remainder and all finite error terms. They are a separate effectivity calculation, not part of the decimal certification.

A Small examples and useful background

A.1 A complete two-by-two system

Take $m = 1, n = 4$, with intervals $[0, 4]$ and $[1, 3]$. Here $d_4 = 12$, the interval lengths are $4, 2$, and

$$C_n = 1! \cdot 12 \cdot 4! \cdot 2! = 576.$$

The exact coefficient matrix is

$$A_{1,4}(x) = \begin{bmatrix} 4x^4 + 112x^3 - 112x - 4 & -48x^3 - 144x^2 - 48x \\ -160x^3 - 144x^2 + 288x + 16 & 48x^3 + 288x^2 + 144x \end{bmatrix}.$$

Its determinant is

$$\det A_{1,4}(x) = 192x(x-1)^6.$$

The exponent $6 = 4 + 2$ is N_n, while the factor x is $x^{(4-4)+(4-3)}$. The two remainders are

$$R_{0,4}(x) = A_{0,0,4}(x) + A_{0,1,4}(x) \log x, \quad R_{1,4}(x) = A_{1,0,4}(x) + A_{1,1,4}(x) \log x.$$

Their first possible orders at $x = 1$ are respectively seven and six, in agreement with the residue-at-infinity calculation. At $x = \mathrm{i}$ the determinant is nonzero.

This example detects a normalization error immediately. Using $d_4/(4!2!) = 1/4$ instead of 576 would divide the whole matrix by 2304. Its x^4 coefficient in the upper-left entry would be $1/576$, rather than the integer 4.

A.2 Why the minimal polynomial cannot be skipped

Consider $y = \mathrm{i}$ and $U(Y) = Y^2 + \mathrm{i}Y - 1$. Then $U(\mathrm{i}) = -3 \neq 0$, but

$$Y^3 + \mathrm{i} = (Y - \mathrm{i})(Y^2 + \mathrm{i}Y - 1).$$

The product of U over all three roots of $Y^3 + \mathrm{i}$ is zero. Nonvanishing at one root alone would not justify a nonzero full-binomial norm. The minimal polynomial of y over $\mathbb{Q}(\mathrm{i})$ is $Y - \mathrm{i}$; its resultant with U is -3. The same logical issue arises for perfect-power rational approximations. Lemma 5.1 handles it uniformly.

A.3 Divisibility before and after evaluation

A prime dividing every coefficient of a polynomial also divides its value at i in $\mathbb{Z}[\mathrm{i}]$. The converse fails. Evaluation can create cancellations, and an individual p-adic unit coefficient need not remain visible in the sum. For example, $P(x) = 1 + x^2$ has unit coefficients at every odd prime, but $P(\mathrm{i}) = 0$. Less extreme cancellations can raise valuations without making the value zero. Thus boundary coefficient calculations alone cannot bound the gcd of an evaluated row from above.

Our balanced arc minimum $f = 0$ likewise proves only that the particular coefficientwise divisor has $\gamma = 0$. It is not a proof that every possible evaluated-row divisor has rate zero.

A.4 The least common multiple as an arithmetic cost

Prime factorization gives

$$\log d_n = \sum_{p^a \le n} \log p = n + o(n).$$

The last equality uses the prime number theorem. Thus d_n^m has exponential cost m, although it is much smaller than $(n!)^m$. The interval factorials in C_n, by contrast, cancel the factorial scale of the contour denominator. Their position is crucial before Stirling's formula is applied.

Factors such as $m!$ can be enormous but are fixed as n grows. Their logarithms divided by n vanish. For large m, small-n regressions may confuse these constants with exponential improvement. Exact identities avoid this ambiguity.

A.5 A two-layer arc calculation by hand

For $L = 14, z = 3/40$, the outer zero arc is $[0, \{14z\}] = [0, 1/20]$, and the inner one is $[\{z\}, \{13z\}] = [3/40, 39/40]$. They are disjoint, as drawn in Figure 1. With weights $128, 128$, at least one type contributes 128 at every pole location. Formula (3.6) collects all such intervals and every periodic prime range.

B Finite experiments and further directions

These computations distinguish analytic convergence from arithmetic usefulness. None proves impossibility for its entire method family.

B.1 Scalar Padé approximation to a pure cubic target

The rational-coefficient function

$$F(z) = \frac{\arcsin\sqrt{z}}{\sqrt{z}} = \sum_{j \ge 0} \frac{\binom{2j}{j}}{4^j(2j+1)} z^j$$

satisfies $F(1/4)^3 = \pi^3/27$. An exact diagonal $[n/n]$ Padé approximant to F^3, evaluated at $1/4$ and multiplied by 27, yields a pure π^3 approximation. After complete rational reduction:

n	$\log Q_n/n$	$-\log\lvert Q_n\pi^3 - P_n\rvert/n$
8	37.2732	−32.0818
16	75.4662	−70.2532
32	155.3497	−150.1009

The denominator growth in these samples overwhelms the analytic error reduction, so they do not give shrinking integer linear forms. An approximate quadratic trend in $\log Q_n$ is only an observation.

B.2 A derivative kernel isolating $\beta(3)$

For integers $a \geq 3, n \geq 1$, define

$$\mathcal{R}_n(t) = \frac{(-1)^n (an)!\, 4^{3n}}{(n!)^3} \frac{\prod_{j=0}^{n-1}(t-j)^3}{\prod_{k=0}^{an}(t+k+1/2)}.$$

It is proper with simple poles, and

$$\mathcal{R}_n(t) = \sum_{k=0}^{an} \frac{(-1)^k b_k}{t+k+1/2}, \quad b_k = \binom{an}{k} \left(\frac{(2k+2n)!\, k!}{(2k)!\, (k+n)!\, n!} \right)^3.$$

The factorial ratio is integral: it is $4^n (k+1/2)_n / n!$, the coefficient of z^n in

$$(1-4z)^{-k-1/2} = (1-4z)^{-k}(1-4z)^{-1/2},$$

whose two factors have integral coefficients.

Two derivatives leave only cubic poles. Hence

$$\mathcal{S}_n = \frac{1}{2} \sum_{t=0}^{\infty} (-1)^t \mathcal{R}_n''(t) = q_n \beta(3) - p_n,$$

$$q_n = 8 \sum_{k=0}^{an} b_k, \qquad p_n = \sum_{k=0}^{an} b_k \sum_{\ell=0}^{k-1} \frac{8(-1)^\ell}{(2\ell+1)^3}.$$

The series is absolutely convergent, since $\mathcal{R}_n''(t) = O(t^{-3})$ even at $a = 3$. The triple zeros make the summands at $0, \ldots, n-1$ zero. No $\beta(1)$ or $\beta(2)$ term occurs.

Nevertheless, clearing the actual denominator of p_n and cancelling the complete integer gcd does not make the tested forms small. For $a = 3$:

n	$\log Q_n / n$	$-\log \lvert Q_n \beta(3) - P_n \rvert / n$
8	24.3881	−14.7970
20	25.5856	−15.8243
40	26.4280	−16.7265

The sampled indices up to 40, for $a = 3, \ldots, 8$, all have negative decay indicators. This rejects the initial parameter choices as presently tested, not every related kernel.

B.3 What a successful binary family would have to prove

Suppose a new construction supplies nonzero integer linear forms

$$L_n = Q_n \beta(3) - P_n, \quad |L_n| = e^{-\delta n + o(n)}, \quad |Q_n| \leq e^{\kappa n + o(n)}, \quad \delta > 0.$$

With nonproportionality allowing a determinant comparison at every denominator scale, one obtains $\mu(\beta(3)) \leq 1 + \kappa/\delta$. Targets 100 and 10 correspond to $\delta/\kappa \geq 1/99$ and $\delta/\kappa \geq 1/9$. The derivative experiment isolates the target correctly but has not reached $\delta > 0$.

A next kernel should preserve isolation while improving factorial-ratio divisibility, possibly through related hypergeometric representations. A second route is the fixed-degree polynomial lower bound in Section 6. Both require new arithmetic estimates: a small residual before denominator clearing does not suffice.

B.4 Corrections that should not become unproved claims

A pointwise local-product limit is not a global steepest-descent proof. For example, replacing the balanced G_0 by

$$G_0 + M \log\cos\frac{\pi}{4M}$$

at $m = 289, M = 290$ changes it by only $-0.001063536257139\ldots$. Even if justified, direct substitution gives $2028.126464240039\ldots$, not roughly 2018. No such sharpened oscillatory estimate is used here.

Finite gcd samples also do not prove that an exponential divisor is present or absent. Our main prime-divisor lower bound is proved coefficientwise and does not depend on a gcd regression.

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